

YEAR 7 MATHEMATICS

Chapter 12

Ratios, rates and time

WORKSHEETS

12

How to use this booklet

- 1** There is **one worksheet for each lesson**. It has three columns that get harder from left to right: **Easy**, then **Medium**, then **Challenging**.

- 2** Read the **speech bubble** at the top of each column: it tells you what to do. A new bubble means a new kind of question.

WE DO **Top of each column.** Your teacher works the examples on the board. **Copy every step** into the space.

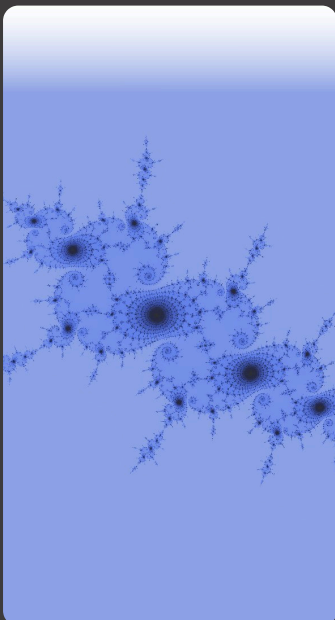
YOU DO **The rest of the column.** Your turn. Column 1: answer on the line. Columns 2 and 3: show your working on the lines.

Contents and progress tracker

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In this chapter you will

- write, simplify and use ratios
- solve ratio problems
- write rates and find the best buy
- solve rate problems and read travel graphs
- round, convert and calculate with time
- use 12-hour and 24-hour time, time differences and timetables



About the artwork

The pictures in this booklet are parts of the **Mandelbrot set**, drawn by a computer that repeats one simple rule, $z \rightarrow z^2 + c$, thousands of times for every point. Every booklet shows a different part of the set: this one is a jellyfish of spirals near the neck of the left bulb. Maths can make beautiful things.

Acknowledgement of Country

We acknowledge the Bundjalung people, the Traditional Custodians of the land on which Kingscliff High School stands, and pay our respects to Elders past and present.

Easy

Questions 1–18

Write the ratio in the order asked. Do not simplify.



WE DO With your teacher: copy the working.

E1 10 green, 6 red: green to red _____

E2 13 yellow, 9 blue: yellow to blue _____

YOU DO Your turn.

1 1 blue, 15 red: red to blue **15 : 1**

2 7 yellow, 7 green: green to yellow _____

3 9 red, 9 yellow: yellow to red _____

4 2 red, 15 yellow: red to all _____

5 14 green, 4 red: red to green _____

6 1 blue, 4 yellow: blue to yellow _____

7 10 yellow, 12 red: yellow to red _____

8 5 green, 10 blue: blue to green _____

9 6 red, 10 yellow: red to all _____

10 5 red, 5 blue: red to blue _____

11 7 blue, 2 yellow: blue to all _____

12 4 yellow, 4 red: yellow to red _____

13 5 yellow, 13 red: yellow to red _____

14 13 blue, 14 red: blue to all _____

15 3 blue, 9 yellow: blue to yellow _____

16 11 green, 2 red: red to green _____

17 2 yellow, 2 red: red to yellow _____

18 8 yellow, 6 blue: yellow to all _____

Medium

Questions 19–28

Write the ratio, then simplify it: divide both parts by the HCF.



WE DO With your teacher: copy the working.

E1 16 girls to 14 boys _____

E2 24 girls to 42 boys _____

YOU DO Your turn.

19 18 girls to 24 boys

18 : 24

÷ 6: 3 : 4

20 20 girls to 35 boys

20 : 35

÷ ____ : ____ : ____

21 25 girls to 40 boys

____ : ____

÷ ____ : ____ : ____

22 14 girls to 6 boys

23 20 girls to 30 boys

24 2 girls to 10 boys

25 36 girls to 12 boys

26 2 girls to 18 boys

27 40 girls to 10 boys

28 20 girls to 25 boys

Challenging

Questions 29–35

Change to the same unit first, then write the ratio in simplest form.



WE DO With your teacher: copy the working.

E1 120 g to 3 kg _____

E2 20c to \$3 _____

YOU DO Your turn.

29 50 min to 4 h

4 h = 240 min

50 : 240

= ____ : ____

30 45 g to 2 kg

____ kg = ____ g

____ : ____

= ____ : ____

31 90 g to 2 kg

32 10 cm to 4 m

33 20 cm to 3 m

34 30 cm to 4 m

35 40 g to 1 kg

12.02 Equivalent and simplified ratios

YEAR 7 · CHAPTER 12: RATIOS, RATES AND TIME

Easy

Questions 1–29

Simplify the ratio: divide both parts by the highest common factor.



WE DO With your teacher: copy the working.

E1 $36 : 27$ _____

E2 $4 : 8$ _____

YOU DO Your turn.

1 $28 : 32$ 7 : 8

2 $16 : 28$ _____

3 $12 : 42$ _____

4 $4 : 2$ _____

5 $18 : 81$ _____

6 $36 : 63$ _____

7 $7 : 35$ _____

8 $72 : 63$ _____

9 $15 : 36$ _____

10 $20 : 15$ _____

11 $42 : 49$ _____

12 $20 : 22$ _____

13 $15 : 3$ _____

14 $35 : 20$ _____

15 $50 : 15$ _____

16 $28 : 8$ _____

17 $12 : 16$ _____

18 $36 : 21$ _____

19 $54 : 24$ _____

20 $20 : 25$ _____

21 $56 : 63$ _____

22 $99 : 90$ _____

23 $88 : 72$ _____

24 $35 : 10$ _____

25 $49 : 70$ _____

26 $10 : 8$ _____

27 $27 : 72$ _____

28 $18 : 16$ _____

29 $9 : 33$ _____

Medium

Questions 30–51

Find the missing number. What was each part multiplied by?



WE DO With your teacher: copy the working.

E1 $3 : 4 = 9 : \square$ _____

E2 $7 : 3 = \square : 18$ _____

YOU DO Your turn.

30 $8 : 1 = \square : 8$

31 $5 : 9 = 40 : \square$

$1 \times 8 = 8$

$5 \times 8 = 40$

$\square = 8 \times 8 = 64$

$\square = ___ \times ___ = ___$

32 $6 : 1 = \square : 4$

33 $1 : 1 = \square : 4$

$___ \times ___ = ___$

$\square = ___ \times ___ = ___$

34 $6 : 3 = 24 : \square$

35 $1 : 7 = \square : 56$

36 $9 : 2 = 18 : \square$

37 $6 : 8 = 24 : \square$

38 $7 : 2 = \square : 12$

39 $9 : 3 = 72 : \square$

40 $4 : 8 = \square : 32$

41 $6 : 9 = \square : 36$

42 $2 : 5 = 10 : \square$

43 $9 : 6 = 36 : \square$

44 $2 : 8 = \square : 56$

45 $7 : 9 = \square : 81$

46 $8 : 9 = 16 : \square$

47 $1 : 7 = \square : 21$

48 $2 : 6 = 8 : \square$

49 $2 : 6 = \square : 24$

50 $3 : 6 = \square : 12$

51 $7 : 5 = \square : 15$

Challenging

Questions 52–61

Simplify. Make whole numbers first ($\times 10$), then divide by the HCF.



WE DO With your teacher: copy the working.

E1 $0.5 : 0.2$ _____

E2 $4 : 32 : 4$ _____

YOU DO Your turn.

52 $28 : 8 : 28$

HCF = 4

$= ___ : ___ : ___$

53 $0.6 : 0.2$

$\times ___ : ___ : ___$

$= ___ : ___$

54 $0.1 : 0.6$ _____

55 $36 : 20 : 20$ _____

56 $36 : 20 : 36$ _____

57 $28 : 12 : 16$ _____

58 $0.2 : 0.8$ _____

59 $8 : 28 : 16$ _____

60 $0.9 : 0.7$ _____

61 $0.8 : 0.4$ _____

Easy

Questions 1–29

Share the amount in the ratio. Add the parts, find one part, then multiply.



WE DO

With your teacher: copy the working.

E1 \$60 in 4 : 1

E2 \$42 in 3 : 3

YOU DO Your turn.

1 \$45 in 2 : 3

\$18, \$27

2 \$16 in 5 : 3

3 \$70 in 4 : 3

4 \$136 in 4 : 4

5 \$21 in 3 : 4

6 \$36 in 3 : 1

7 \$80 in 3 : 1

8 \$72 in 3 : 1

9 \$30 in 1 : 2

10 \$52 in 1 : 3

11 \$36 in 2 : 2

12 \$110 in 5 : 5

13 \$18 in 3 : 3

14 \$65 in 2 : 3

15 \$40 in 2 : 2

16 \$14 in 3 : 4

17 \$152 in 5 : 3

18 \$28 in 3 : 1

19 \$70 in 4 : 1

20 \$28 in 1 : 1

21 \$84 in 2 : 4

22 \$112 in 4 : 3

23 \$55 in 3 : 2

24 \$40 in 1 : 4

25 \$56 in 3 : 5

26 \$63 in 4 : 3

27 \$36 in 3 : 3

28 \$60 in 3 : 2

29 \$60 in 3 : 3

Medium

Questions 30–36



Share in the ratio. Show the total parts and one part.

WE DO

With your teacher: copy the working.

E1 \$34 in the ratio 1 : 1

E2 \$50 in the ratio 5 : 5

YOU DO Your turn.

30 \$136 in the ratio 4 : 4

parts: $4 + 4 = 8$ one part: $136 \div 8 = 17$

\$68 and \$68

31 \$60 in the ratio 1 : 5

parts: $1 + 5 = 6$ one part: $60 \div 6 = 10$

___ and ___

32 \$35 in the ratio 1 : 4

parts: ___ + ___ = ___

one part: ___ $\div$ ___ = ___

___ and ___

33 \$150 in the ratio 5 : 5

34 \$42 in the ratio 1 : 2

35 \$88 in the ratio 5 : 3

36 \$63 in the ratio 4 : 5

Challenging

Questions 37–41

Use the ratio to find the missing amount.



WE DO

With your teacher: copy the working.

E1 Cordial : water = 4 : 8. How much water for 280 mL of cordial?

E2 Boys : girls = 3 : 8. There are 64 girls. How many students altogether?

YOU DO Your turn.

37 Cordial : water = 5 : 7. How much water for 150 mL of cordial?

5 parts = 150 mL

1 part = 30 mL

water: ___ $\times$ ___ = ___ mL

38 Boys : girls = 3 : 2. There are 8 girls. How many students altogether?

___ parts = ___, ___ part = ___

boys: ___ $\times$ ___ = ___

total: ___ + ___ = ___

39 Cordial : water = 3 : 8. How much water for 90 mL of cordial?

40 Boys : girls = 6 : 5. There are 50 girls. How many students altogether?

41 Cordial : water = 6 : 4. How much water for 540 mL of cordial?

Easy

Questions 1–30

Write as a rate for one unit (divide).

**WE DO** With your teacher: copy the working.

E1 273 m in 7 s

E2 \$188 in 4 kg

YOU DO Your turn.29
words/min

1 116 words in 4 min

2 \$208 in 4 kg

3 290 m in 5 s

4 200 words in 4 min

5 180 words in 6 min

6 100 L in 5 min

7 297 m in 9 s

8 114 km in 6 h

9 368 m in 8 s

10 \$343 in 7 kg

11 \$287 in 7 kg

12 \$301 in 7 kg

13 207 m in 9 s

14 57 m in 3 s

15 182 km in 7 h

16 75 m in 5 s

17 \$164 in 4 kg

18 252 words in 9 min

19 378 L in 7 min

20 78 m in 6 s

21 348 L in 6 min

22 120 m in 3 s

23 36 m in 4 s

24 472 L in 8 min

25 384 km in 8 h

26 \$54 in 9 kg

27 24 words in 2 min

28 144 m in 9 s

29 212 km in 4 h

30 18 km in 2 h

Medium

Questions 31–38

Write the rate, then divide to find the rate for one unit.

**WE DO** With your teacher: copy the working.E1 A car travels 525 km in 5 hours.
Find its speed.E2 A car travels 872 km in 8 hours.
Find its speed.**YOU DO** Your turn.31 A car travels 300 km in 4 hours.
Find its speed. $300 \text{ km} \div 4 \text{ h}$ $= 75 \text{ km/h}$ 32 A car travels 808 km in 8 hours.
Find its speed. $808 \text{ km} \div 8 \text{ h}$ $= \text{ } \text{ km/h}$ 33 A car travels 742 km in 7 hours.
Find its speed. $\text{ } \text{ km} \div \text{ } \text{ h}$ $= \text{ } \text{ km/h}$ 34 A car travels 288 km in 3 hours.
Find its speed.35 A car travels 470 km in 5 hours.
Find its speed.36 A car travels 138 km in 3 hours.
Find its speed.37 A car travels 552 km in 6 hours.
Find its speed.38 A car travels 427 km in 7 hours.
Find its speed.

Challenging

Questions 39–44

Change both rates to the same unit, then compare.

**WE DO** With your teacher: copy the working.

E1 Car A: 35 km/h. Car B: 142 km in 3 h. Which is faster?

E2 Car A: 36 km/h. Car B: 355 km in 4 h. Which is faster?

YOU DO Your turn.

39 Car A: 74 km/h. Car B: 109 km in 4 h. Which is faster?

B: $109 \div 4 = 27.3 \text{ km/h}$

A: 74 km/h

faster: $\text{ } \text{ km/h}$

40 Car A: 47 km/h. Car B: 214 km in 4 h. Which is faster?

B: $\text{ } \div \text{ } = \text{ } \text{ km/h}$ A: $\text{ } \text{ km/h}$ faster: $\text{ } \text{ km/h}$

41 Car A: 41 km/h. Car B: 258 km in 4 h. Which is faster?

42 Car A: 55 km/h. Car B: 399 km in 4 h. Which is faster?

43 Car A: 56 km/h. Car B: 240 km in 3 h. Which is faster?

44 Car A: 46 km/h. Car B: 104 km in 2 h. Which is faster?

Easy

Questions 1–30

Find the unit price: the price of one.

**WE DO**

With your teacher: copy the working.

E1 9 for \$74.16

E2 6 for \$8.52

YOU DO Your turn.

1 10 for \$47.90

\$4.79

2 11 for \$79.97

3 8 for \$55.84

4 7 for \$41.09

5 4 for \$21.44

6 7 for \$38.43

7 6 for \$34.80

8 6 for \$31.02

9 3 for \$9.99

10 6 for \$18.90

11 11 for \$36.52

12 2 for \$4.32

13 9 for \$47.88

14 4 for \$10.32

15 2 for \$16.14

16 9 for \$77.94

17 2 for \$5.98

18 7 for \$44.38

19 12 for \$50.28

20 3 for \$6.78

21 4 for \$24.28

22 9 for \$68.40

23 7 for \$54.04

24 4 for \$20.60

25 4 for \$23.00

26 8 for \$6.32

27 6 for \$30.96

28 3 for \$2.52

29 10 for \$23.90

30 7 for \$17.92

Medium

Questions 31–40

Find the price of one of each, then choose the better buy.

**WE DO**

With your teacher: copy the working.

E1 2 for \$6.16 or 4 for \$11.00

E2 6 for \$6.60 or 5 for \$11.35

YOU DO Your turn.

31 3 for \$10.20 or 2 for \$3.14

\$10.20 ÷ 3 = \$3.40; \$3.14 ÷ 2 = \$1.57

better: 2 for \$3.14

32 3 for \$5.88 or 6 for \$13.74

\$5.88 ÷ 3 = \$1.96; \$13.74 ÷ 6 = \$2.29

better: ___ for ___

33 5 for \$15.75 or 3 for \$10.98

___ ÷ ___ = ___; ___ ÷ ___ = ___

better: ___ for ___

34 5 for \$15.05 or 6 for \$22.92

35 2 for \$4.90 or 3 for \$10.86

36 3 for \$10.77 or 4 for \$7.12

37 2 for \$7.50 or 5 for \$16.55

38 3 for \$6.60 or 5 for \$14.30

39 6 for \$23.52 or 2 for \$5.98

40 3 for \$7.05 or 6 for \$20.16

Challenging

Questions 41–46

Compare the price for 100 g of each. Which is the better buy?

**WE DO**

With your teacher: copy the working.

E1 300 g for \$4.97 or 1000 g for \$11.92

E2 250 g for \$5.71 or 500 g for \$6.52

YOU DO Your turn.

41 300 g for \$2.12 or 1000 g for \$4.14

300 g: \$2.12 ÷ 3 = \$0.71 per 100 g

1000 g: \$4.14 ÷ 10 = \$0.41 per 100 g

better buy: ___ g

42 400 g for \$4.44 or 1000 g for \$5.79

___ g: ___ ÷ ___ = ___ per ___ g

___ g: ___ ÷ ___ = ___ per ___ g

better buy: ___ g

43 200 g for \$1.57 or 500 g for \$5.26

44 400 g for \$2.41 or 1000 g for \$9.94

45 400 g for \$5.53 or 1000 g for \$5.75

46 300 g for \$4.56 or 1000 g for \$10.43

Easy

Questions 1–30

Use the rate to find the total:
rate $\times$ amount.



WE DO With your teacher: copy the working.

E1 10 kg at \$5/kg

E2 7 h at 10 km/h

YOU DO Your turn.

1 8 h at 40 km/h

320 km

2 3 h at 45 km/h

3 11 h at 35 km/h

4 4 h at \$25/h

5 6 h at \$19/h

6 4 kg at \$7/kg

7 7 kg at \$3/kg

8 10 h at \$14/h

9 10 kg at \$24/kg

10 7 kg at \$16/kg

11 10 h at \$22/h

12 6 h at 120 km/h

13 9 h at \$10/h

14 11 h at \$20/h

15 2 kg at \$12/kg

16 10 kg at \$14/kg

17 2 kg at \$14/kg

18 3 kg at \$17/kg

19 4 h at 60 km/h

20 4 kg at \$13/kg

21 3 h at \$21/h

22 8 h at \$9/h

23 9 h at \$2/h

24 10 kg at \$19/kg

25 6 kg at \$16/kg

26 12 h at \$12/h

27 3 h at 115 km/h

28 12 h at 75 km/h

29 11 h at 120 km/h

30 4 h at 40 km/h

Medium

Questions 31–40



Find the time taken: time =
distance $\div$ speed.

WE DO With your teacher: copy the working.

E1 300 km at 60 km/h

E2 120 km at 40 km/h

YOU DO Your turn.

31 160 km at 40 km/h

160 $\div$ 40

= 4 h

32 400 km at 100 km/h

400 $\div$ 100

= ___ h

33 600 km at 100 km/h

___ $\div$ ___

= ___ h

34 100 km at 50 km/h

35 360 km at 60 km/h

36 100 km at 100 km/h

37 300 km at 100 km/h

38 200 km at 40 km/h

39 300 km at 50 km/h

40 50 km at 50 km/h

Challenging

Questions 41–45

Two steps: use the rate
twice. Show each step.



WE DO With your teacher: copy the working.

E1 Mia earns \$16 an hour. She works 7 hours a day for 4 days. How much does she earn?

E2 Mia earns \$27 an hour. She works 6 hours a day for 4 days. How much does she earn?

YOU DO Your turn.

41 Mia earns \$17 an hour. She works 5 hours a day for 2 days. How much does she earn?

one day: 17 $\times$ 5 = \$85

2 days: 85 $\times$ 2

= ___

42 Mia earns \$18 an hour. She works 4 hours a day for 2 days. How much does she earn?

one day: ___ $\times$ ___ = ___

___ days: ___ $\times$ ___

= ___

43 Mia earns \$20 an hour. She works 6 hours a day for 5 days. How much does she earn?

44 Mia earns \$16 an hour. She works 5 hours a day for 3 days. How much does she earn?

45 Mia earns \$17 an hour. She works 8 hours a day for 4 days. How much does she earn?

Easy

Questions 1–30

Find the speed: $\text{speed} = \text{distance} \div \text{time}$.

WE DO With your teacher: copy the working.

E1 120 km in 3 h _____

E2 120 km in 6 h _____

YOU DO Your turn.

1 180 km in 4 h 45 km/h

2 60 km in 3 h _____

3 240 km in 6 h _____

4 40 km in 2 h _____

5 480 km in 6 h _____

6 225 km in 5 h _____

7 360 km in 6 h _____

8 270 km in 6 h _____

9 200 km in 2 h _____

10 90 km in 6 h _____

11 40 km in 4 h _____

12 270 km in 3 h _____

13 400 km in 5 h _____

14 180 km in 2 h _____

15 80 km in 4 h _____

16 30 km in 3 h _____

17 50 km in 5 h _____

18 160 km in 2 h _____

19 80 km in 2 h _____

20 450 km in 5 h _____

21 120 km in 2 h _____

22 60 km in 6 h _____

23 60 km in 4 h _____

24 160 km in 4 h _____

25 400 km in 4 h _____

26 320 km in 4 h _____

27 100 km in 5 h _____

28 45 km in 3 h _____

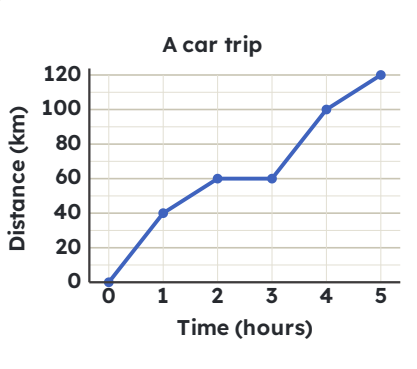
29 135 km in 3 h _____

30 540 km in 6 h _____

Medium

Questions 31–34

Use the travel graph. Read the distance at each hour.



WE DO With your teacher: copy the working.

E1 How far in the first hour? _____

E2 What was the total distance? _____

YOU DO Your turn.

31 When did the car stop?

the flat part of the line

= from 2 h to 3 h

32 Speed in the 2nd hour?

$60 - 40 = 20 \text{ km in } 1 \text{ h}$

= ____ km/h

33 Speed in the 4th hour?

____ - ____ = ____ km in ____ h

= ____ km/h

34 Average speed for the trip?

Challenging

Questions 35–40

Find the missing value: $d = s \times t$, $s = d \div t$, $t = d \div s$.

WE DO With your teacher: copy the working.

E1 $s = 50 \text{ km/h}$, $t = 4 \text{ h}$. Find d .

E2 $d = 80 \text{ km}$, $t = 4 \text{ h}$. Find s .

E3 $d = 250 \text{ km}$, $s = 50 \text{ km/h}$. Find t .

YOU DO Your turn.

35 $s = 60 \text{ km/h}$, $t = 4 \text{ h}$. Find d .

$d = s \times t$

$= 60 \times 4$

$= \text{____ km}$

36 $d = 240 \text{ km}$, $t = 4 \text{ h}$. Find s .

$= \text{____} \div \text{____}$

$= \text{____ km/h}$

37 $d = 240 \text{ km}$, $s = 60 \text{ km/h}$. Find t .

38 $s = 40 \text{ km/h}$, $t = 3 \text{ h}$. Find d .

39 $d = 200 \text{ km}$, $t = 4 \text{ h}$. Find s .

40 $d = 120 \text{ km}$, $s = 60 \text{ km/h}$. Find t .

Easy

Questions 1–22

Convert. 1 h = 60 min, 1 min = 60 s.



WE DO With your teacher: copy the working.

E1 420 min = ____ h

E2 6 min = ____ s

YOU DO Your turn.

1 7 h = ____ min 420 min

2 6 h = ____ min

3 120 min = ____ h

4 3 min = ____ s

5 5 h = ____ min

6 4 min = ____ s

7 4 h = ____ min

8 8 min = ____ s

9 300 min = ____ h

10 2 min = ____ s

11 2 h = ____ min

12 360 min = ____ h

13 8 h = ____ min

14 480 min = ____ h

15 5 min = ____ s

16 7 min = ____ s

17 9 h = ____ min

18 240 min = ____ h

19 9 min = ____ s

20 540 min = ____ h

21 180 min = ____ h

22 3 h = ____ min

Medium

Questions 23–32

Add the times. Add hours and minutes separately, then change 60 min to 1 h.

WE DO With your teacher: copy the working.

E1 3 h 46 min + 2 h 15 min

E2 3 h 50 min + 2 h 20 min

YOU DO Your turn.

23 1 h 33 min + 1 h 53 min

2 h 86 min

= 3 h 26 min

24 2 h 26 min + 1 h 28 min

3 h 54 min

= ____ h ____ min

25 3 h 24 min + 2 h 30 min

____ h ____ min

= ____ h ____ min

26 3 h 20 min + 2 h 30 min

27 2 h 18 min + 2 h 50 min

28 3 h 15 min + 1 h 23 min

29 3 h 30 min + 1 h 12 min

30 3 h 44 min + 2 h 24 min

31 1 h 12 min + 2 h 35 min

32 2 h 45 min + 2 h 45 min

Challenging

Questions 33–42

Change between decimal hours and hours and minutes.

WE DO With your teacher: copy the working.

E1 2 h 45 min = ____ h (decimal)

E2 5 h 24 min = ____ h (decimal)

YOU DO Your turn.

33 3.4 h = ____ h ____ min

$0.4 \times 60 = 24$ min

= ____ h ____ min

34 6 h 6 min = ____ h (decimal)

____ min ÷ ____ = ____

= ____ h

35 4 h 12 min = ____ h (decimal)

36 1 h 45 min = ____ h (decimal)

37 3.5 h = ____ h ____ min

38 6.1 h = ____ h ____ min

39 4 h 30 min = ____ h (decimal)

40 6 h 12 min = ____ h (decimal)

41 4 h 15 min = ____ h (decimal)

42 5 h 30 min = ____ h (decimal)

Easy

Questions 1–25

Write in 24-hour time. For p.m. times, add 12 to the hour.

WE DO With your teacher: copy the working.

E1 10:25 p.m. _____

E2 4:05 a.m. _____

YOU DO Your turn.

1 12:00 p.m. 12:00

2 2:35 a.m. _____

3 11:10 a.m. _____

4 8:40 a.m. _____

5 3:20 p.m. _____

6 2:00 a.m. _____

7 6:55 a.m. _____

8 4:00 p.m. _____

9 12:10 a.m. _____

10 3:40 p.m. _____

11 4:55 p.m. _____

12 1:15 p.m. _____

Write in 12-hour time with a.m. or p.m.

WE DO With your teacher: copy the working.

E3 14:00 _____

YOU DO Your turn.

13 12:50 12:50 p.m.

14 03:45 _____

15 15:20 _____

16 17:45 _____

17 21:25 _____

18 12:10 _____

19 20:15 _____

20 15:45 _____

21 08:15 _____

22 05:00 _____

23 19:05 _____

24 14:10 _____

25 14:05 _____

Medium

Questions 26–35

Convert. Take care near midnight and midday.

WE DO With your teacher: copy the working.

E1 11:40 p.m. _____

E2 12:55 p.m. _____

YOU DO Your turn.

26 12:15 p.m.

p.m.: 12 p.m. stays 12

= 12:15

27 11:15 p.m.

p.m.: add 12

= ____:____

28 12:05 a.m.

a.m.: ____ a.m. is ____

= ____:____

29 12:30 a.m. _____

30 11:55 p.m. _____

31 12:35 p.m. _____

32 11:20 p.m. _____

33 12:50 p.m. _____

34 11:40 a.m. _____

35 12:40 a.m. _____

Challenging

Questions 36–42

Find the time after the minutes given. Write it in both forms.

WE DO With your teacher: copy the working.

E1 80 min after 08:15 _____

E2 50 min after 21:55 _____

YOU DO Your turn.

36 15 min after 14:05

14:05 + 15 min

= 14:20

= ____:____ p.m.

37 60 min after 11:50

____:____ + ____ min

= ____:____

= ____:____ p.m.

38 115 min after 03:00 _____

39 110 min after 13:15 _____

40 20 min after 12:25 _____

41 135 min after 04:45 _____

42 60 min after 16:05 _____

Easy

Questions 1–18

Find the time difference.
Count up to the next hour,
then on.



WE DO

With your teacher: copy the
working.

- E1 1:10 p.m. to 2:40 p.m. _____
- E2 9:20 a.m. to 10:30 a.m. _____

YOU DO Your turn.

- 1 11:15 a.m. to 2:20 p.m. 3 h 5 min
- 2 12:50 p.m. to 2:15 p.m. _____
- 3 9:10 a.m. to 12:45 p.m. _____
- 4 8:35 a.m. to 1:35 p.m. _____
- 5 7:20 a.m. to 9:20 a.m. _____
- 6 1:05 p.m. to 1:40 p.m. _____
- 7 10:10 a.m. to 11:10 a.m. _____
- 8 11:15 a.m. to 1:05 p.m. _____
- 9 9:10 a.m. to 11:45 a.m. _____
- 10 1:05 p.m. to 5:45 p.m. _____
- 11 1:15 p.m. to 4:10 p.m. _____
- 12 8:25 a.m. to 9:30 a.m. _____
- 13 11:45 a.m. to 2:45 p.m. _____
- 14 10:55 a.m. to 11:25 a.m. _____
- 15 7:55 a.m. to 9:00 a.m. _____
- 16 7:30 a.m. to 8:15 a.m. _____
- 17 10:00 a.m. to 11:15 a.m. _____
- 18 12:05 p.m. to 2:50 p.m. _____

Medium

Questions 19–25

Count up in steps: to the
next hour, then whole hours,
then the minutes.



WE DO

With your teacher: copy the
working.

- E1 16:25 to 21:30

- E2 15:30 to 18:45

YOU DO Your turn.

- 19 08:05 to 09:45
08:05 → 09:00: 55 min
09:00 → 09:45: 45 min
total: 1 h 40 min
- 20 11:25 to 13:00
11:25 → 12:00: 35 min
12:00 → 13:00: 1 h
total: ___ h ___ min
- 21 15:50 to 18:00
___:___ → ___:___: ___ min
___:___ → ___:___: ___ h
total: ___ h ___ min
- 22 12:55 to 14:15

- 23 07:55 to 11:45

- 24 15:35 to 20:00

- 25 10:35 to 14:20

Challenging

Questions 26–29

Time zones: Sydney is 2
hours ahead of Perth. Find
the time, then the flight
arrival.



WE DO

With your teacher: copy the
working.

- E1 A flight leaves Perth at 2:35 p.m. and takes 4 h 50 min. What is the time in Sydney when it lands?

- E2 A flight leaves Perth at 11:35 a.m. and takes 3 h 40 min. What is the time in Sydney when it lands?

YOU DO Your turn.

- 26 A flight leaves Perth at 3:00 p.m. and takes 3 h 55 min. What is the time in Sydney when it lands?
lands: 3:00 p.m. + 3 h 55 min = 6:55 p.m. Perth
+ 2 h for Sydney
= ___:___ p.m.
- 27 A flight leaves Perth at 7:30 a.m. and takes 3 h 30 min. What is the time in Sydney when it lands?
lands: ___:___ a.m. + ___ h ___ min = ___:___
+ ___ h for Sydney
= ___:___ p.m.
- 28 A flight leaves Perth at 11:00 a.m. and takes 4 h 5 min. What is the time in Sydney when it lands?

- 29 A flight leaves Perth at 7:10 a.m. and takes 4 h 5 min. What is the time in Sydney when it lands?

Easy

Questions 1–14

Use the bus timetable. Read across for the stop and down for the bus.

Stop	Bus 1	Bus 2	Bus 3	Bus 4
School	08:05	08:35	09:20	09:35
Library	08:12	08:42	09:27	09:42
Pool	08:24	08:54	09:39	09:54
Shops	08:33	09:03	09:48	10:03
Beach	08:47	09:17	10:02	10:17

WE DO With your teacher: copy the working.

E1 Bus 1 at the Library _____

E2 Bus 2 at the Library _____

YOU DO Your turn.

1 Bus 3 at the Library 09:27

2 Bus 4 at the Library _____

3 Bus 1 at the Pool _____

4 Bus 2 at the Pool _____

5 Bus 3 at the Pool _____

6 Bus 4 at the Pool _____

7 Bus 1 at the Shops _____

8 Bus 2 at the Shops _____

9 Bus 3 at the Shops _____

10 Bus 4 at the Shops _____

11 Bus 1 at the Beach _____

12 Bus 2 at the Beach _____

13 Bus 3 at the Beach _____

14 Bus 4 at the Beach _____

Medium

Questions 15–24

How long is the trip? Subtract the times from the timetable.

WE DO With your teacher: copy the working.

E1 Bus 1: School to Library _____

E2 Bus 2: Library to Shops _____

YOU DO Your turn.

15 Bus 3: Pool to Shops

09:39 to 09:48

= 9 min

16 Bus 4: School to Pool

09:35 to 09:54

= ____ min

17 Bus 1: Library to Pool

____:____ to ____:____

= ____ min

18 Bus 2: Pool to Beach

19 Bus 3: School to Library

20 Bus 4: Library to Shops

21 Bus 1: Pool to Shops

22 Bus 2: School to Pool

23 Bus 3: Library to Pool

24 Bus 4: Pool to Beach

Challenging

Questions 25–27

Plan a trip with the timetable. Give reasons.

WE DO With your teacher: copy the working.

E1 You must be at the Beach by 10:07. What is the latest bus you can catch from School?

E2 You reach the Library at 08:44. When is the next bus, and when does it reach the Pool?

YOU DO Your turn.

25 How long is the whole route from School to Beach?

08:05 to 08:47

= ____ min

26 Write Bus 4's time at the Shops in 12-hour time.

____:____

= ____:____ a.m.

27 How many minutes apart do Bus 1 and Bus 2 leave School?